Gaoxiang Zhao

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Education

University of Pennsylvania — MSE in Robotics, Philadelphia, USA Aug 2025 – May 2027

GPA 3.95/4.00 · Machine Learning (A), Learning in Robotics (A), Physical Intelligence (A)

Wuhan University — BEng in Communication Engineering, Wuhan, China Sep 2021 – Jul 2025

Experience

GRASP Lab, University of Pennsylvania — Research Assistant Dec 2025 – Present

- Current research on humanoid robot learning: whole-body control policies learned from human demonstration data, with imitation learning, reinforcement learning and world models.
- Earlier work on event-based vision: built a synthetic data generation pipeline with domain randomization and photorealistic rendering for event-camera object pose estimation.

Tencent — Research Intern, Shenzhen, China Jun 2026 – Aug 2026

- Worked on agentic 3D scene generation and 3D asset generation.

Manycore Tech — Research Intern, Hangzhou, China Aug 2024 – Nov 2024

- Developed Monte Carlo physically based rendering algorithms for a production-scale rendering pipeline supporting photorealistic synthetic data generation.

Selected Projects

Humanoid Motion Tracking on the Unitree G1 Sep 2026 – Present

- Built a pipeline from human motion capture or a single monocular video to physically tracked motion on the Unitree G1: retargeting with GMR, audits on every clip, and two tracking backends scored by one evaluation protocol, all in simulation.
- Trained a multi-motion PPO tracking policy in Isaac Lab that tracks 190 of 230 held-out test clips, and reproduced SuperMimic, a world-model tracking policy, which completes 195 of 230.
- Extended the tracking to highly dynamic acrobatics, such as cartwheels and back handsprings, retargeted from motion capture under a whole-body floor constraint.

Sim-to-Real Drone Racing with Deep RL Feb 2026 – Apr 2026

- Trained a PPO racing policy in Isaac Lab across 8,192 parallel environments, with dynamics domain randomization and curriculum learning; a simplified adaptation of Kaufmann et al. (*Nature* 2023).
- Transferred it zero-shot to a real Crazyflie 2.1, which flew 3 laps of a 7-gate course in 20 s.

Physically Based Rendering & Kinematics in C++ 2023 – 2025

- A renderer from scratch with path tracing, bidirectional path tracing, photon mapping and multiple importance sampling, plus a kinematics library with three IK solvers (analytical, CCD, Jacobian pseudoinverse).
- Implemented the free-space diffraction BSDF of Steinberg et al. (2024) in PBRT-v3, adding a spectral rendering mode.

Technical Skills

Programming: C++, Python, MATLAB

Tools: PyTorch, Isaac Lab, Isaac Sim, MuJoCo, ROS 2, OpenGL, PBRT, Linux, Git, Slurm

Languages: English (proficient), Mandarin (native)